

ideally replace all battery back-up solutions and enable an environment-friendly product for its customers," Kumar explains.

LED-based lighting products are highly efficient in terms of energy requirement in domestic and industrial segments.

"STMICROELECTRONICS has the best-in-class power supply controllers enabling optimised energy efficiency. Similarly, low-power microcontrollers are available with high processing capabilities. This opens the opportunities for many applications which can have low power consumption and hence lower carbon footprint," shares Kumar.

According to Kumar, brushless DC motor based fans consume almost half of the power consumed by split-phase motor based ceiling fans generally available in India. This is achieved by the field-oriented control library running on a microcontroller in combination with a small low-loss intelligent moulded module (SLIMM). This provides new opportunities in power saving and enables domestic appliances to offer better energy ratings.

Compared to a conventional driver, which typically requires more than 30 individual components, SLIMMs reduce costs by simplifying design and significantly reducing component count while saving space, improving reliability and lowering electromagnetic emissions.

Additional cost incurred?

"The cost of using RoHS-compliant material instead of components that contain lead is minimally additional. RoHS-certified PCBs are now manufactured in India. These can cost higher than the legacy PCB manufacturing technology. RoHS-compliant solder too is now easily available in India. Here again, using lead-free solder is costlier and may need different handling of the assembly of electronic boards," informs Kumar.

Furthermore, if a firm chooses

Hazardous chemicals in electronic products

1. Lead can be found in solders, although decreasingly, in the glass of cathode ray tube (CRT) monitors and as a stabiliser in PVC. It is highly toxic and exposure to lead can result in irreversible damage to the nervous system, particularly in children, which can lead to intellectual impairment.

2. Mercury, used in lighting devices for most flat-screen displays, can damage the brain and central nervous system, particularly during early development.

3. Cadmium, used in rechargeable computer batteries, contacts and switches and in older CRTs, can accumulate in the body over time and is highly toxic, primarily affecting the kidneys and bones. Cadmium and its compounds are also known human carcinogens.

4. Beryllium, used as a metal alloy in electrical contacts and as beryllium oxide in the semiconductor industry, is a human carcinogen and inhalation of fumes and dusts can cause lung disease.

5. Compounds of hexavalent chromium, used in the production of metal housings, are highly toxic and human carcinogens.

6. Some BFRs used in circuit boards and plastic casings do not break down easily and can build up in the environment, and some BFRs are also highly bio-accumulative (build up in the body). Long-term exposure to certain polybrominated diphenylethers has been linked to abnormal brain development in animals, with possible impacts on learning, memory and behaviour. Some BFRs can also interfere with thyroid and oestrogen hormone systems and exposure in the womb has been linked to behavioural problems. Incineration or any kind of burning of plastics containing BFRs can cause the release of persistent dioxins and furans.

7. PVC is a chlorinated plastic used in some electronics products, including for insulation on wires and cables. Although not directly toxic, it is a major source of pollution and chemical hazard at all stages of its life cycle. In its softened form (as found in cables), PVC requires the use of additives such as hazardous phthalates, including di(2-ethylhexyl) phthalate and di-n-butyl phthalate, which are known as reproductive toxins. Incineration or any kind of burning of PVC can cause the release of persistent and toxic chlorinated dioxins and furans.

Source: 'Switching On To Green Electronics' report by Greenpeace

to make its just one or two product lines green, the cost is higher. If it makes all its product lines green, the additional cost is only marginal or negligible. In the case of computers, for example, there is only 0.5 per cent additional cost incurred because these are not manufactured in hundreds or thousands of units but hundreds of thousands of units.

By taking the green considerations into all stages of the product life cycle, including R&D, design, procurement, production, sale, logistics, use and disposal, manufacturers can, in fact, attain cost saving and benefit customers of their product offerings. Renesas claims to have been able to do so.

"Our RX MCU, which is embedded with industry-leading eFlash FMONOS process, is currently the fastest in the industry with a

10ns max read access time and is designed for optimised power consumption up to 100 MHz operations. In order to reduce cost and power consumption, we have migrated RX MCU to 40nm process. Furthermore, by making the MONOS process platform available to other semiconductor suppliers with the collaboration recently announced with TSMC, our FMONOS platform benefit could reach a wider customer base," informs Ronnie Ho.

"Making a green product is not a philanthropic act but a business strategy as it gives your money back or, in other words, maximises your net profits. It also sustains your business. In electronics, if the margin is less than 2 per cent, you can't afford to keep spending money on certain things which don't give you big returns," comments Pratap.